



Engineering Modern AI 2026: From Foundation Models to Multimodal AI

A Two-Day Hands-on Workshop on Training Modern Large Language Models and Vision Language Models

About the Workshop

Artificial Intelligence has rapidly evolved from deep learning systems into frontier foundation models capable of reasoning, coding, multimodal understanding, and autonomous decision-making. This workshop provides a comprehensive overview of the modern AI landscape, exploring the breakthroughs, architectures, and training methodologies that power today's state-of-the-art language and vision models.

Participants will learn how modern AI systems are built from large-scale pretraining and post-training to reinforcement learning, test-time scaling, evaluation, and multimodal learning. Through hands-on sessions, they will set up a cloud-based training environment and train both a reasoning Large Language Model (LLM) and a Vision Language Model (VLM) using contemporary industry techniques.

Learning Outcomes

- Understand the historical evolution of modern AI
- Explain how Large Language Models are pretrained and post-trained
- Understand scaling laws, datasets, GPUs, and training infrastructure
- Learn how reasoning models such as DeepSeek-R1 are created
- Understand Reinforcement Learning for LLMs
- Learn Test-Time Scaling methodologies
- Fine-tune their own reasoning language model
- Build and train a Vision Language Model
- Understand the emerging AI ecosystem including agents and multimodal systems
- Explore future research directions and career opportunities in AI

Day 1

Session 1 — Evolution of Modern AI (40 mins)

- What is Intelligence?
- Defining Artificial Intelligence
- Measuring Intelligence
- Brief History of AI- ImageNet, AlexNet, DeepMind, AlphaGo, AlphaZero, AlphaFold
- Prominent People in AI
- Rise of Transformers
- GPT-2, GPT-3
- Dawn of reasoning models- OpenAI o1, o3
- ARC-AGI Benchmark
- The DeepSeek Moment
- Current AI Landscape

Session 2 — AI/ML Fundamentals (20 mins)

- Machine Learning
- Neural Networks
- Backpropagation

Session 3 — Inside Large Language Models (1 hour)

- Transformer Architecture - Attention is all you Need
- Tokenization
- Pretraining, Post-Training, RLHF
- Scaling Laws
- Data Pipelines
- GPU and CUDA
- Hyperscalers
- Model Evaluation
- Distillation
- Test-Time Scaling - Reasoning Models

Session 4 — Reinforcement Learning for LLMs (45 mins)

- Introduction to RL
- RLHF, RLAIIF
- PPO, DPO
- GRPO
- Reward Models
- Chain of Thought
- Cold Start SFT
- Reasoning Data Generation

Hands-on Lab 1 (15 mins)

Setting up the AI Training Environment

- Introduction to Jupyter Notebook
- Colab, Kaggle and Modal

Hands-on Lab 2 (1.5 hours)

Training a Reasoning Language Model

- Model Selection
- Prepare instruction datasets
- Perform Cold-Start SFT
- Apply GRPO-based Reinforcement Learning
- Evaluate model performance
- Compare reasoning improvements

Day 2

Session 4 — The Modern AI Ecosystem (30 mins)

- Foundation Models
- Open Source vs Closed Models
- AI Agents - Tools Use, Function Calling, MCP, Skill Files, Products
- Current AI Market
- Frontier AI Companies

Session 5 — Vision Language Models (35 mins)

- Why Multimodal AI?
- Vision Encoders
- Projectors
- Cross-Modal Alignment
- VLM Architectures
- Modern Vision Models

Hands-on Lab 3 (1 hour)

Training a Vision Language Model

- Collecting image-text datasets
- Pretraining vision projector
- Training Vision SFT
- Vision GRPO
- Evaluate multimodal performance

Session 6 — The Future of AI (40 mins)

- Scaling Beyond LLMs
- Agentic Systems
- Scientific AI
- Robotics
- World Models
- Physics-Informed Machine Learning
- AI Research Trends
- Career Opportunities
- Open Problems in AI

Thank You!

Prerequisites

- Basic Python Programming
- Linear Algebra Fundamentals
- Calculus Fundamentals

Tools & Frameworks

- Python
- PyTorch

- Hugging Face Transformers
- TRL
- Unsloth
- Modal
- vLLM
- Cloud GPUs